

1(a)	<i>any one from:</i> volume of (pH7) buffer ; speed of centrifuge ; time in centrifuge ;	1
1(b)	<i>idea that</i> time for disc rising decreases as catalase activity increases ; ora	1
1(c)(i)	distance from base of shoot ;	1
1(c)(ii)	<i>any six from:</i> 1 method to cut, tissue from carrot (root) ; 2 minimum of 5 distances chosen (or stated distances) ; 3 same / stated, mass / volume, of cut carrot (used to make the extract) ; 4 mix / stir, liquid carrot extract (before dipping filter paper discs) ; 5 method of maintaining temperature (10°C – 40°C if stated) ; 6 same / stated, height / depth / (stated) volume, of hydrogen peroxide (solution) ; 7 measure / note / record, the time the filter paper disc takes to rise to the surface (of hydrogen peroxide solution) ; 8 use at least three measurements for each distance and calculate a mean ; 9 safety: hazard and risk and mitigation ; 10 carrying out a control with disc, dipped / AW, in water / buffer (to show rise of the discs is due to the carrot extract) ;	6
1(d)(i)	volume of gas collected (after 1 minute) ;	1

Question	Answer	Marks
1(d)(ii)	distilled water / deionised water / water, in table heading ; correct volumes of stock solution and water (in Table 1.1) ;	2
1(d)(iii)	<i>supports conclusions:</i> 1 paired data quotes / manipulated data (from Fig. 1.7), to show increase in hydrogen peroxide concentration increases the volume of gas collected ; <i>doesn't support – max 2 marks:</i> 2 <i>idea that</i> range (of hydrogen peroxide concentrations) is limited ; 3 <i>idea that</i> graph has not, plateaued / reached a maximum / AW, so we do not know the value of $V_{\max}$; 4 <i>idea that</i> gas collected might not be only oxygen ; 5 no replicates / no repeats of investigation / no mean calculated ;	3

Question	Answer	Marks
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